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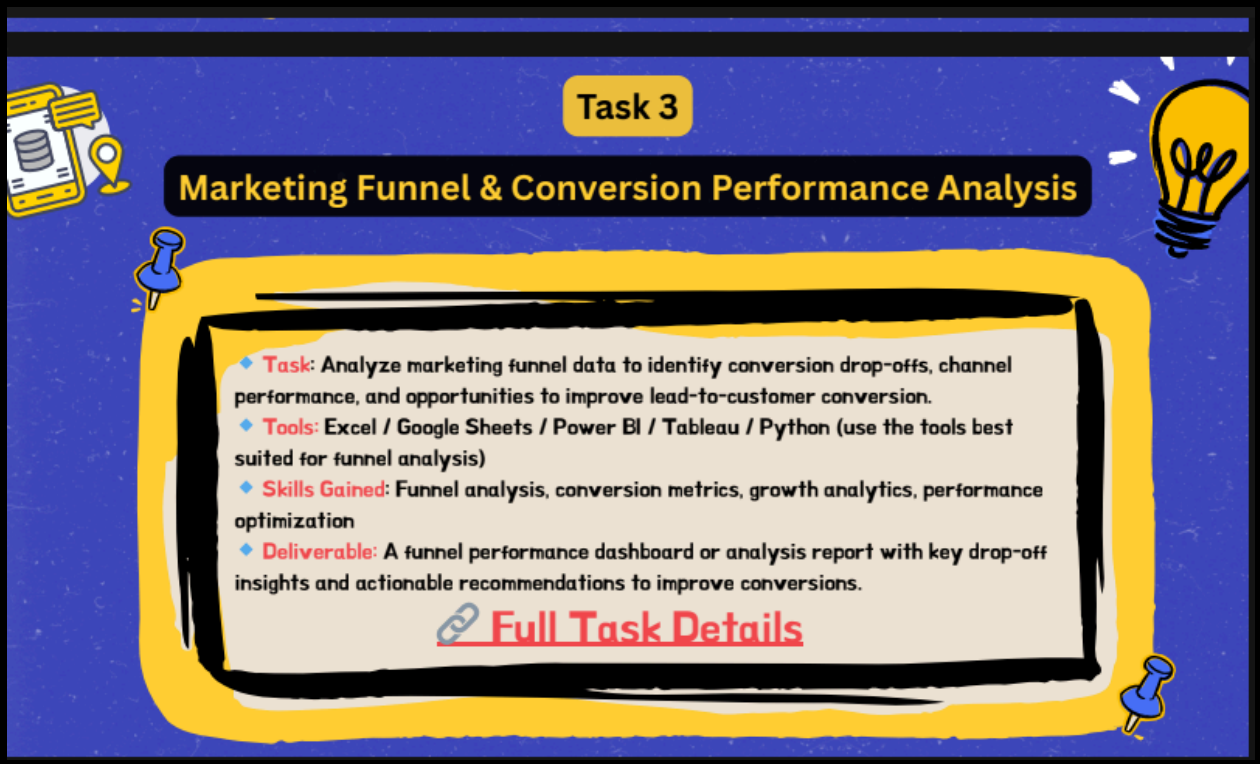
CONVERSION PERFORMANCE ANALYSIS



About the Task

In this task, we will analyze marketing and lead funnel data to understand how users move from visitors → leads → customers.

This type of analysis is core work in startups, digital marketing agencies, SaaS companies, and sales teams, where improving conversion rates directly increases revenue.

Your goal is to help a business answer questions like:

- Where are users dropping off in the funnel?
- Which channels bring high-quality leads?
- How can conversion rates be improved?
- Which stages need optimization?

What we have done

- Clean and structure funnel or lead data
- Analyze:
    - Traffic-to-lead conversion
    - Lead-to-customer conversion
    - Drop-off points at each funnel stage
- Compare performance across channels, campaigns, or time periods
- Provide clear recommendations to improve conversions

Tools we used

- Google Sheets
- Google Docs
- Power BI

Skills I Gained

- Funnel and conversion analysis
- Marketing & growth analytics
- KPI tracking and performance metrics
- Business-focused insight generation

- Data-driven decision making

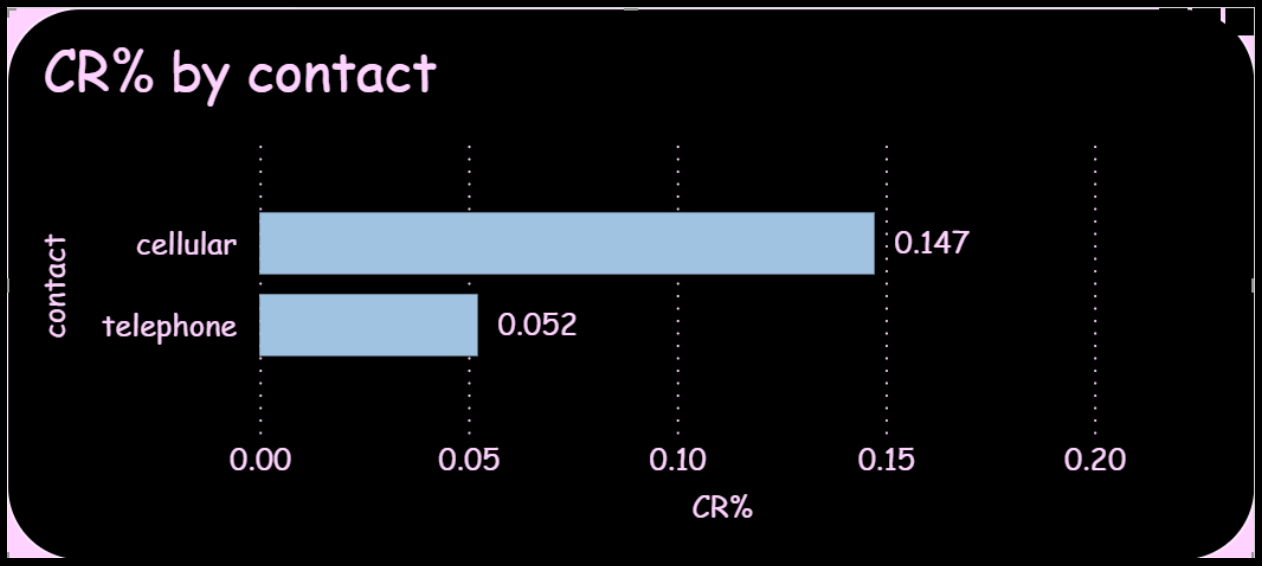
Analytical Study

Key Performance Indicators (KPIs)

41188	4640	11.27%
Total Leads	Total Conversions	CR%

- **Campaign Conversion Count ('yes'):**
  The total number of clients who successfully subscribed to the term deposit.
- **Campaign Rejection Count ('no'):**
  The total number of clients who declined the term deposit.
- **Total Market Reach (Volume):**
  The combined total of both bars (the entire sample size of the campaign).
- **Subscription Success Rate (Class Balance):**
  The visual ratio between the orange and blue bars, showing the proportion of "wins" versus "losses."

Horizontal Bar Chart

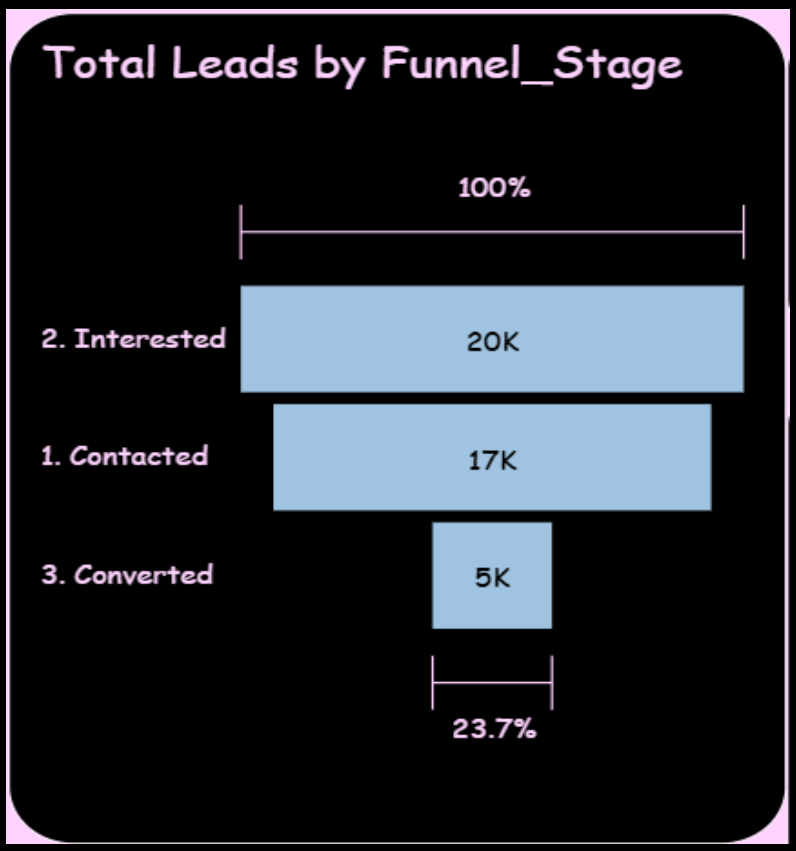


This chart shows the **Conversion Rate (CR%) based on contact type**.

There are two contact methods in the dataset:

- Cellular
  - Conversion Rate = 0.147
    This means 14.7% of customers contacted via cellular converted.
- Telephone
  - Conversion Rate = 0.052
    This means 5.2% of customers contacted via telephone converted.

Conversion Funnel Chart



The chart shows **Total Leads by Funnel Stage** and how leads move through different stages:

Contacted – 17K

- 17,000 leads were initially contacted.
- This represents the starting stage of the funnel.

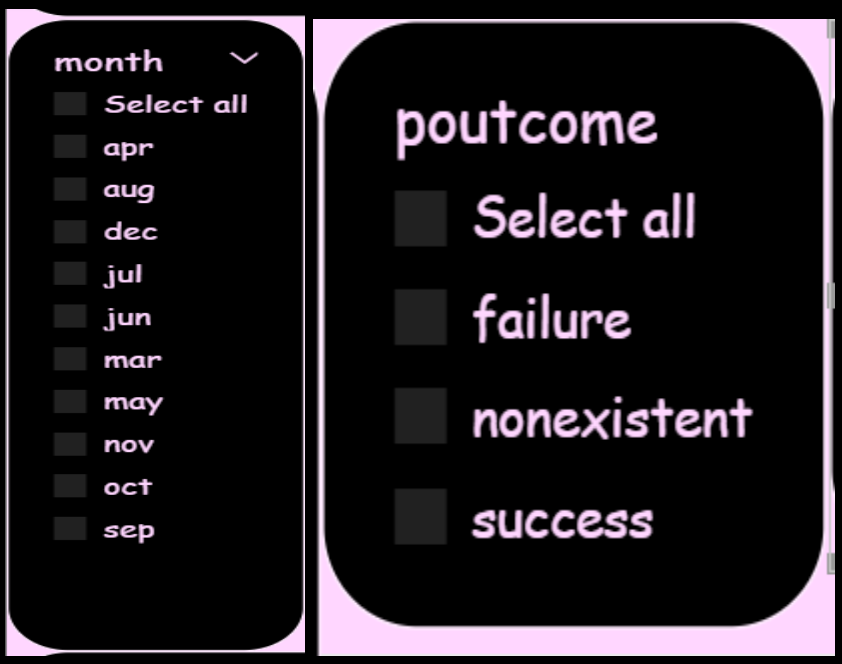
Interested – 20K

- 20,000 leads are marked as interested.
- This is shown as **100%** in the chart (top reference level).

Converted – 5K

- 5,000 leads finally converted.
- This represents **23.7% conversion rate** from the total interested leads.

Checkbox Slicers



1. Month Slicer:

This slicer contains months:

- Apr
- Aug
- Dec
- Jul
- Jun
- Mar
- May
- Nov
- Oct
- sep

Purpose:

- It filters the data based on the **month in which customers were contacted**.
- When a specific month is selected, all charts (like CR%, funnel, etc.) will show data only for that selected month.

- "Select all" option shows data for all months.

Insight Use:

This helps analyze:

- Which month had higher leads
- Which month had better conversion rate
- Seasonal trends in campaign performance

2. Poutcome Slicer

It contains:

- failure
- nonexistent
- success

Meaning in Dataset:

“Poutcome” stands for **Previous Campaign Outcome**.

- **success** → Customer responded positively in previous campaign
- **failure** → Customer did not respond
- **nonexistent** → Customer was not contacted in previous campaign

Purpose:

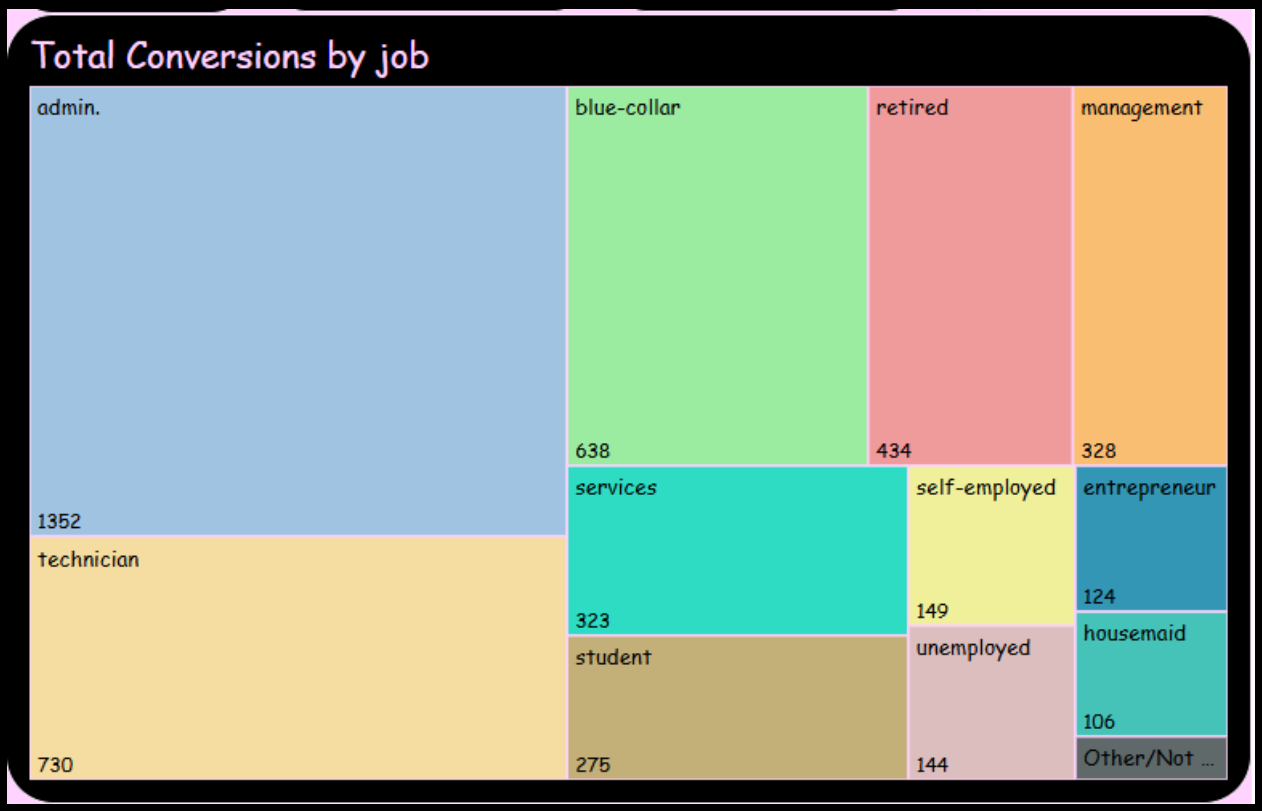
- Filters data based on previous campaign result.
- Helps analyze whether past campaign success affects current conversion.

Insight Use:

This helps determine:

- Do previously successful customers convert again?
- Does previous failure impact new campaign results?

Treemap Analysis



- Market Share (Rectangle Size):

  The area of each rectangle represents the **Volume** of a specific category (e.g., `job` or `education`). Larger boxes indicate the most frequently targeted or occurring groups in the dataset (e.g., "blue-collar" or "admin.").

- **Conversion Success (Color Intensity):**

  The color usually represents the **Subscription Rate** (KPI: % of 'yes' outcomes). A darker or distinct color highlights which segments are most profitable, regardless of their size.

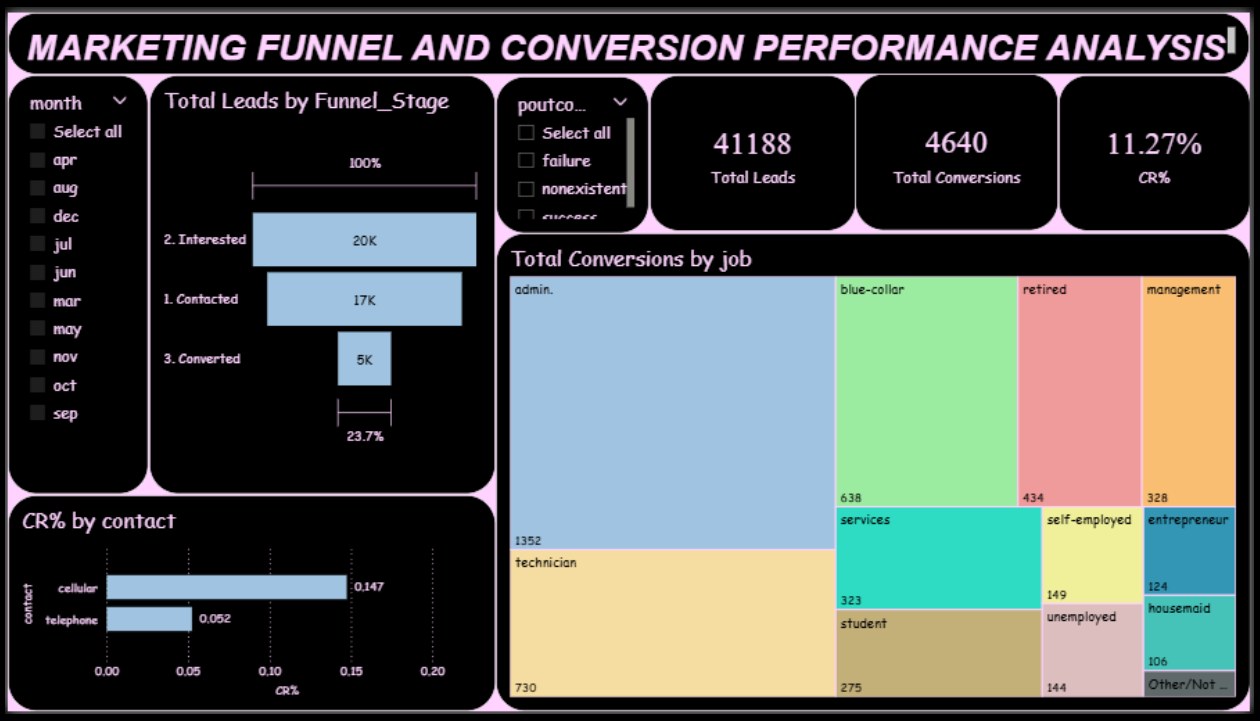
- **Categorical Composition:**

  If the map is grouped (e.g., `marital` status inside `job` types), it shows the sub-demographics of your most active leads.

- **Outreach Distribution:**

  It allows for an instant audit of where the bank's resources are concentrated versus where they are seeing the best results.

Dashboard Overview



Based on your data, here is how to explain the specific components:

1. The Bar Chart (Class Distribution)

- **KPI: Lead Volume vs. Success.**
- **Explanation:** This shows the massive imbalance in your dataset. The "no" bar represents the ~88% of clients who declined, while the "yes" bar represents the ~12% conversion rate.
- **Insight:** It sets the "Baseline" for the campaign's success.

2. The Clustered Bar Chart (Segmented Analysis)

- **KPI: Conversion Rate by Demographic.**
- **Explanation:** This breaks down the results by a specific attribute (e.g., **Job**). It compares "yes" and "no" side-by-side for each group.

- **Insight:** It helps identify which groups are "high-effort" (many "no"s, like blue-collar) versus "high-efficiency" (proportionally more "yes"s, like students or retired).

3. The Treemap (Market Composition)

- **KPI: Targeting Concentration.**
- **Explanation:** The size of the boxes shows which customer segments dominate your dataset (e.g., "Married" or "University Degree").
- **Insight:** If the largest boxes (most calls) don't have the highest success color, it suggests the bank is targeting the wrong audience.

4. The Confusion Matrix (Model Accuracy)

- **KPI: Predictive Reliability.**
- **Explanation:** This evaluates how well a model can predict the "yes" responses.
    - **True Positives:** Successfully identifying a subscriber.
    - **False Positives:** Calling someone who isn't interested (waste of resources).
- **Insight:** It measures the "Technical Success" of the data science effort.

<u>Summary for your report:</u>

"This dashboard provides a 360-degree view of the telemarketing campaign. It transitions from **Total Volume** (Bar Chart) to **Targeting Strategy** (Treemap and Clustered Chart), and finally to **Predictive Performance** (Confusion Matrix). Collectively, these KPIs help the bank reduce 'Wasted Outreach' and focus on segments with the highest 'Subscription Probability'."

Visual Insights

Based on the **Bank Marketing dataset**, here are the key visual insights summarized for your analysis:

- Class Imbalance:

  A bar chart of the target variable y shows a significant skew, with ~88% "no" and ~12% "yes." This indicates that the campaign has a low overall conversion rate.

- **High-Performing Segments:**

  Clustered bar charts by `job` or `education` reveal that while "admin" has the highest volume, "students" and "retired" groups show the highest proportional success rates.

- **Call Duration Impact:**

  Box plots show a strong positive correlation between `duration` (call length) and success, suggesting that longer engagements are much more likely to result in a subscription.

- **Economic Influence:**

  Analysis of the `euribor3m` and `cons.conf.idx` fields shows that subscriptions often decrease during periods of high interest rates or low consumer confidence.

- Targeting Concentration:

  A treemap by `marital` or `housing` status reveals that the bank focuses heavily on married individuals with housing loans, though this may not always be the most efficient segment for conversions.

- **Model Accuracy:**

  A confusion matrix highlights the difficulty in predicting the minority "yes" class, showing where the model typically confuses interested leads with uninterested ones.

Actionable Recommendations (Where to grow?

Based on the patterns found in the Bank Marketing dataset, here are the short, actionable recommendations on where to focus your growth strategy:

- **Target "High-Efficiency" Demographics:**

  Shift focus toward **Students** and **Retired** clients. While they represent a smaller volume of the data, they consistently show the highest conversion rates compared to the saturated "blue-collar" and "admin" segments.

- **Prioritize "Previous Success" Leads:**

  Clients who subscribed in previous campaigns (where poutcome = "success") are significantly more likely to subscribe again. Create a "Loyalty" track for these individuals.

- **Optimize Call Timing:**

  Data shows higher success rates in specific months (like **March, September, and October**). Increasing campaign intensity during these "high-conversion" windows rather than the high-volume month of May can improve ROI.

- **Leverage Economic Windows:**

  Monitor the **Euribor 3-month rate**. Conversion rates tend to be higher when interest rates are lower. Launch aggressive growth campaigns when the economic environment indicates high consumer confidence.

- **Focus on Single & Highly Educated Leads:**

  Clients with a **University Degree** or those who are **Single** have a higher propensity to subscribe to term deposits. Refine marketing messaging to appeal to the financial goals of these specific profiles.

- **The "Quality over Quantity" Call Strategy:**

  Since duration is a high predictor of success, train agents on "Engagement Depth" rather than "Call Volume." A single 5-minute call is statistically more valuable than five 1-minute calls.